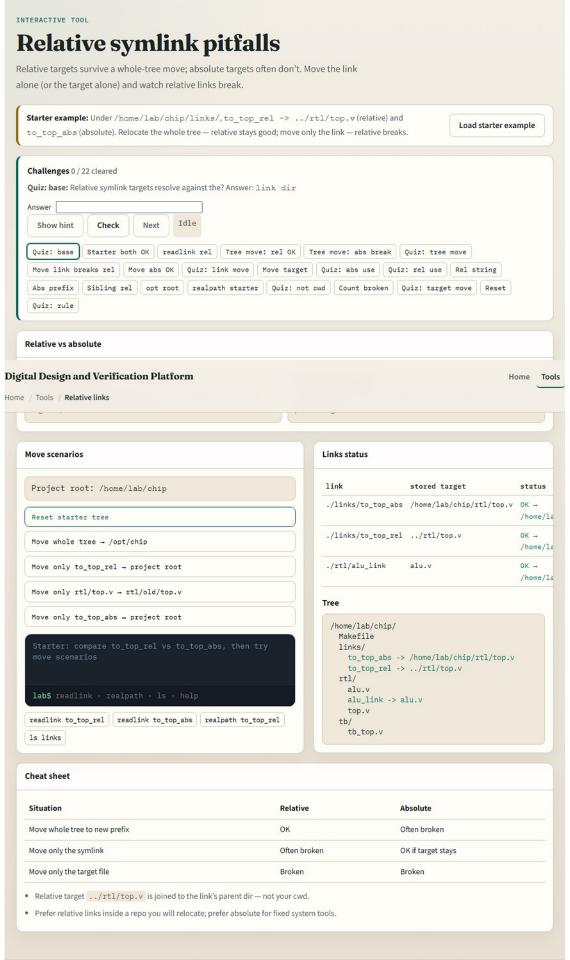
Relative symlink pitfalls

Symlinks let a project point at shared IP without copying files

Relative to the link, not your cwd

- The target string is resolved from the directory that contains the symlink
- From project slash external, shared lib is up two levels then shared slash lib
- Prefer relative inside a repo you will move; prefer absolute only for fixed system tools
- Moving only the symlink to a new folder often breaks a relative link

Browser lab



Link relative lab

Real shell practice

```
wsl — module22-link-relative/examples/link_relative

# Track A — real Unix

# real shell session (Track A)

$ ls -la external/
total 0
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 .
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 ..
-rwxrwxrwx 1 yongfu yongfu    0 Jul 18 16:34 .gitkeep
lrwxrwxrwx 1 yongfu yongfu   18 Jul 18 16:34 lib.v -> ../../shared/lib.v

$ readlink external/lib.v
../../shared/lib.v

$ cat external/lib.v
// Shared library for link_relative example
module lib;
endmodule

$ ln -s "$(realpath ../shared/lib.v)" external/abs_lib.v

$ ls -la external/
total 0
drwxrwxrwx 1 yongfu yongfu 4096 Jul 20  2026 .
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 ..
-rwxrwxrwx 1 yongfu yongfu    0 Jul 18 16:34 .gitkeep
lrwxrwxrwx 1 yongfu yongfu  106 Jul 20  2026 abs_lib.v -> /mnt/d/proj/designs/learning/courses/learn_unix/module22-link-
relative/examples/link_relative/shared/lib.v
lrwxrwxrwx 1 yongfu yongfu   18 Jul 18 16:34 lib.v -> ../../shared/lib.v

$ readlink external/abs_lib.v
/mnt/d/proj/designs/learning/courses/learn_unix/module22-link-relative/examples/link_relative/shared/lib.v

$ rm external/abs_lib.v
```

Real shell — relative link to shared IP

Real shell practice — try these

```
# ls -la external/ — see lib.v -> ../../shared/lib.v
ls -la external/

# readlink external/lib.v — print the stored relative target
readlink external/lib.v

# cat external/lib.v — follow the link and show shared content
cat external/lib.v

# ln -s "$(realpath …)" — absolute contrast (fixed path; breaks if
tree
moves)
ln -s "$(realpath ../shared/lib.v)" external/abs_lib.v

readlink external/abs_lib.v
```

Pitfalls to watch

- Count the dots from the link's directory, not from the repo root by habit
- Do not move a relative symlink alone without updating its target string
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, run whole-tree relocate and move-only-link
- On the real shell, inspect the external lib link with readlink and cat
- When you are ready